Xiao FU (付潇)

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Education

The Chinese University of Hong Kong (CUHK) — Hong Kong SAR
Ph.D. in Information Engineering, Multimedia Laboratory (MMLab) — *2023 – Present*
- **Advisor:** Prof. *Dahua Lin*

Zhejiang University (ZJU) — Hangzhou, China
B.Eng. (Honor) in Information Engineering — *2018 – 2022*
- **Advisor:** Prof. *Yiyi Liao*, **GPA:** 3.96/4.00 (Top 1%)

Industrial Research

- **Has GPT-6-Astra Achieved Embodied Intelligence?**
  Xiao Fu (Contributor)
  SenseNova Team White Paper, 2026 | **website** | **package** | **code**
- **Cosmos-Interactive: Interactive World Simulation with Cosmos for Physical AI**
  Xiao Fu (Core Contributor): Model Post-training, Model Ablation, Data Preparation, Benchmarks
  NVIDIA Cosmos White Paper, 2026 | **website** | **product** | **code**
- **Cosmos 3: Omnimodal World Models for Physical AI**
  Xiao Fu (Core Contributor): Long-Context Interactivity, Data Curation, Synthetic Data Generation
  NVIDIA Cosmos White Paper, 2026 | **website** | **product** | **code (12K+ stars)**
- **Cosmos 2.5: World Simulation with Video Foundation Models for Physical AI**
  Xiao Fu (Core Contributor): Multi-Camera & Interactive Video Generation, Multi-View Robotic Transfer
  NVIDIA Cosmos White Paper, 2025 | **website** | **code (predict2.5)** | **code (transfer2.5)**

Selected Publications

- **Plenoptic Video Generation**
  Xiao Fu, Shitao Tang, Min Shi, Xian Liu, Jinwei Gu, Ming-Yu Liu, Dahua Lin, Chen-Hsuan Lin
  Computer Vision and Pattern Recognition (**CVPR**), 2026 | **website** | **code**
- **Learning Video Generation for Robotic Manipulation with Collaborative Trajectory Control**
  Xiao Fu, Xintao Wang, Xian Liu, Jianhong Bai, Runsen Xu, Pengfei Wan, Di Zhang, Dahua Lin
  International Conference on Learning Representations (**ICLR**), 2026 | **website** | **code (100+ stars)**
- **3DTrajMaster: Mastering 3D Trajectory for Multi-Entity Motion in Video Generation**
  Xiao Fu, Xian Liu, Xintao Wang, Sida Peng, Menghan Xia, Xiaoyu Shi, Ziyang Yuan, Pengfei Wan, Di Zhang, Dahua Lin
  International Conference on Learning Representations (**ICLR**), 2025 | **website** | **code (300+ stars)**
- **PanopticNeRF-360: Panoramic 3D-to-2D Label Transfer in Urban Scenes**
  Xiao Fu, Shangzhan Zhang, Tianrun Chen, Yichong Lu, Xiaowei Zhou, Andreas Geiger, Yiyi Liao
  IEEE Transactions on Pattern Analysis and Machine Intelligence (**T-PAMI**), 2025 | **website** | **code (200+ stars)**
- **GeoWizard: Unleashing the Diffusion Priors for 3D Geometry Estimation from a Single Image**
  Xiao Fu, Wei Yin, Mu Hu, Kaixuan Wang, Yuexin Ma, Ping Tan, Shaojie Shen, Dahua Lin, Xiaoxiao Long
  European Conference on Computer Vision (**ECCV**), 2024 | **website** | **code (900+ stars)** (**Citation: 300+**)
- **Panoptic NeRF: 3D-to-2D Label Transfer for Panoptic Urban Scene Segmentation**
  Xiao Fu, Shangzhan Zhang, Tianrun Chen, Yichong Lu, Lanyun Zhu, Xiaowei Zhou, Andreas Geiger, Yiyi Liao
  International Conference on 3D Vision (**3DV**), 2022 | **website** | **code (200+ stars)** (**Citation: 200+**)
- **ReCamMaster: Camera-Controlled Generative Rendering from A Single Video**
  Jianhong Bai, Menghan Xia, **Xiao Fu**, Xintao Wang, Lianrui Mu, Jinwen Cao, Zuozhu Liu, Haoji Hu, Xiang Bai, Pengfei Wan, Di Zhang
  International Conference on Computer Vision (**ICCV**), 2025 (**Best Paper Award Candidate, Top 13/11,239**)
- **OmniObject3D: Large-Vocabulary 3D Object Dataset for Realistic Perception, Reconstruction and Generation**
  Tong Wu, Jiarui Zhang, **Xiao Fu**, Yuxin Wang, Jiawei Ren, Liang Pan, Wayne Wu, Lei Yang, Jiaqi Wang, Chen Qian, Dahua Lin, Ziwei Liu
  Computer Vision and Pattern Recognition (**CVPR**), 2023 (**Best Paper Award Candidate, Top 12/9,155**)

Work Experience

Cosmos Lab, NVIDIA — Singapore
Research Scientist. Advisor: Dr. *Ming-Yu Liu*, Dr. *Chen-Hsuan Lin* — Jun. 2026 – Aug. 2026
Topic: Cosmos World Foundation Models for Physical AI (Cosmos 3)

Cosmos Lab, NVIDIA — Santa Clara, United States
Research Scientist Intern. Advisor: Dr. *Ming-Yu Liu*, Dr. *Chen-Hsuan Lin* — May 2025 – May 2026
Topic: Cosmos World Foundation Models for Physical AI (Cosmos 2.5 & 3)

Kling Team, Kuaishou VGI Lab — Shenzhen, China
Research Intern. Advisor: Dr. *Xintao Wang* — May 2024 – May 2025
Topic: Controllable Video Generation

Shanghai AI Innovation Center — Shanghai, China
Associate Researcher. Advisor: Prof. *Dahua Lin*, Dr. *Jiaqi Wang* — Jun. 2022 – Jun. 2023
Topic: OmniObject3D: A Large-Vocabulary and Realistic 3D Object Dataset

Max Planck Institute for Intelligent Systems — Tübingen, Germany (remote)
Research Assistant. Advisor: Prof. *Andreas Geiger*, Prof. *Yiyi Liao* — Sept. 2021 – Mar. 2022
Topic: Panoptic Neural Radiance Field for 3D-2D Label Transfer

Invited Talks & Workshops

- **NVIDIA Research**: Multi-Camera Video Generation for Embodied Intelligence *2025*
- **AnySyn3D Webinar**: 3D Interactive Video Generation *2024*
- **AI for 3D Content Creation Workshop at ICCV 2023**: OmniObject3D Challenge *2023*
- **Shenlan Open Courses**: Panoramic Label Rendering for Autonomous Driving Simulator *2023*

Honors & Awards

- ICCV Best Paper Award Candidate *2025*
- CVPR Best Paper Award Candidate *2023*
- Hong Kong PhD Fellowship Scheme (HKPFS) **(Top 300 Ph.D. students in Hong Kong)** *2023*
- CUHK Vice-Chancellor HKPFS Scholarship *2023*
- Outstanding Graduation Thesis Award of Zhejiang University *2022*
- National Scholarship, Ministry of Education of P.R. China **(Top 0.2% across China)** *2020,2021*
- ZJU ISEE Excellent Student Award **(Dean's Honor Undergraduate)** *2021*
- ZJU ISEE Yuelun Alumni Scholarship **(Dean's Award for Academic Contest)** *2020*
- Finalist Winner, Mathematical Contest in Modeling (MCM/ICM) **(Top 2% out of 20,954 teams)** *2020*
- Gold Prize, 7th "Internet Plus" College Student Innovation and Entrepreneurship Contest **(1st author)** *2021*